

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Brijesh.T/192521391 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Unified Big Data Architecture for Retail Customer Analytics

A multinational retail organization can integrate fragmented online and offline customer data using a **unified big data architecture**. Data from e-commerce websites, mobile apps, physical stores, POS systems, CRM, loyalty programs, and social media can be collected through streaming and batch ingestion tools such as **Apache Kafka** and ETL pipelines. The data can be stored in a centralized data lake using cloud storage or Hadoop. **Apache Spark** can process large volumes of data for real-time analytics, while machine learning models analyze customer behavior and generate personalized product recommendations. A unified customer ID can connect interactions across all channels. This architecture provides a complete customer view, improves personalization, increases customer satisfaction, and supports better business decisions.

2. Distributed Computing for Scientific Research Data

Distributed computing is highly effective for processing large-scale scientific data because it divides complex computational tasks among multiple computers or nodes. Frameworks such as **Hadoop**, **Apache Spark**, and high-performance computing clusters allow large datasets to be processed simultaneously. Parallel processing improves computational efficiency by reducing execution time and distributing workloads across available resources. Unlike centralized systems, distributed systems can scale horizontally by adding more machines when data volume increases. However, challenges include network communication, synchronization, and fault management. Overall, distributed computing provides greater scalability, speed, and resource utilization for scientific simulations and data analysis.

3. Big Data Fraud Detection Framework for Digital Payments

A fraud detection framework can collect real-time payment data such as transaction amount, location, device information, transaction frequency, IP address, and user behavior. **Apache Kafka** can capture streaming transactions, while **Apache Spark Streaming** or similar technologies can process them immediately. Historical transaction data can be stored in a data lake for analysis and model training. Predictive machine learning models can identify unusual patterns and assign a fraud risk score to each transaction. High-risk transactions can be blocked or sent for verification. Streaming analytics enables immediate detection, while predictive models improve accuracy by learning from previous fraudulent and legitimate transactions. This reduces financial losses and improves fraud prevention.

4. Managing Multimedia Data in Big Data Environments

Multimedia data such as videos, images, and audio creates challenges because of its large size, different formats, high storage requirements, and complex processing needs. Traditional relational databases are generally not suitable for managing massive multimedia datasets. A suitable solution is to use distributed storage technologies such as **HDFS**, cloud object storage, or distributed file systems. Metadata can be stored separately in databases such as **MongoDB** or other NoSQL systems. Distributed processing frameworks like **Apache Spark** can process metadata and large-scale data, while specialized AI models can analyze images, audio, and videos. Compression, content delivery networks, indexing, and scalable cloud storage improve storage efficiency and retrieval performance.

5. Disaster Recovery and Business Continuity for Big Data Platforms

A cloud-based big data platform should implement a disaster recovery strategy using **redundancy, replication, backup, and automated failover**. Critical data should be replicated across multiple availability zones or geographical regions. Regular backups should be stored separately to protect against accidental deletion or corruption. Multiple processing nodes should be available so workloads can continue if one node fails. Automated failover mechanisms can redirect applications and users to healthy systems without significant interruption. Regular disaster recovery testing should verify that the recovery procedures work correctly. These mechanisms improve availability, reduce downtime, prevent data loss, and ensure business continuity during system failures or disasters.

6. Data Quality and Data Governance in Analytics Platforms

Data quality issues such as inconsistency, duplication, missing values, and incorrect formats can significantly reduce the reliability of analytics results. Duplicate customer records may produce inaccurate customer counts, while missing values can affect machine learning models and statistical analysis. A strong data governance strategy should define data standards, validation rules, ownership, access controls, and quality monitoring. Data cleansing processes can remove duplicates, standardize formats, and handle missing values. A master data management

system can maintain consistent versions of important business entities. Regular data quality audits and automated validation improve the accuracy and reliability of the analytics platform, leading to better decision-making.

7. Scalable Recommendation System for an Online Streaming Service

A scalable recommendation system can analyze user behavior such as viewing history, search activity, watch time, ratings, likes, and skipped content. This data can be collected continuously using streaming technologies and stored in a distributed big data platform. **Apache Spark** can process large-scale user and content data, while machine learning algorithms can identify similarities between users and content. Techniques such as collaborative filtering and content-based recommendation can generate personalized suggestions. Distributed processing allows millions of users and items to be analyzed efficiently. Better analysis of user behavior improves recommendation accuracy, increases user engagement, and helps retain subscribers.

8. Role of Cloud Computing in Big Data Ecosystems

Cloud computing plays a major role in modern big data ecosystems by providing scalable computing, storage, and analytics resources on demand. Organizations can increase or decrease resources based on workload requirements, making cloud platforms highly flexible. Cloud computing reduces the need for expensive physical infrastructure and supports distributed big data tools and services. However, uncontrolled resource usage can increase operational costs. Security, privacy, vendor dependency, and data transfer costs are also important limitations. Therefore, organizations should implement cost monitoring, access control, encryption, backup strategies, and multi-cloud or hybrid-cloud planning. Overall, cloud computing provides significant scalability and flexibility when managed properly.

9. Real-Time Analytics for Transportation and Logistics Management

A real-time transportation analytics framework can collect data from GPS devices, vehicles, traffic systems, weather services, fuel sensors, and delivery applications. Streaming platforms can process this information continuously and send it to a big data processing system. Analytics algorithms can identify traffic congestion, calculate optimal routes, monitor vehicle performance, and predict delivery delays. Machine learning models can also analyze historical and real-time information to improve route planning. Route optimization reduces travel distance and fuel consumption, while real-time monitoring improves operational decision-making. This framework helps logistics organizations reduce costs, improve delivery performance, and increase overall efficiency.

10. Ethical and Legal Challenges of Big Data Analytics

Big data collection creates ethical and legal challenges related to **privacy, consent, data security, transparency, bias, and misuse of personal information**. Organizations may collect more user data than necessary or use it for purposes that users do not clearly understand. Biased datasets can also produce unfair automated decisions. To ensure responsible data usage, organizations should establish clear data governance policies, obtain informed consent, minimize unnecessary data collection, and protect sensitive information through encryption and access controls. Compliance with applicable data protection laws and regulations should be regularly monitored. Organizations should also conduct algorithm audits and provide transparency about how data is collected and used. These policies improve regulatory compliance, protect user rights, and build long-term user trust.